

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL
INTELLIGENCE
AND EXPERT SYSTEMS

COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

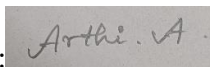
Annexure A

Agent Design Assessment

Code of Conduct:

I Arthi A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

ARTIFICIAL INTELLIGENCE LAB ASSIGNMENT

Title

Development of a Student Attendance Chatbot Using Google Dialogflow

Question 1: Problem Analysis and Agent Design

Problem Analysis

In educational institutions, students frequently ask questions regarding their attendance percentage and attendance records. Answering these queries manually requires faculty members or administrative staff to spend considerable time responding to each student individually. This process can be time-consuming and inefficient.

To solve this problem, an Artificial Intelligence (AI)-based conversational agent can be developed using Google Dialogflow. The chatbot automatically understands student queries written in natural language and provides instant responses about attendance information.

How Artificial Intelligence Solves the Problem

Artificial Intelligence uses **Natural Language Processing (NLP)** to understand user questions, identify the user's intent, extract important information such as the subject name, and generate an appropriate response.

For example:

User: "What's my attendance in DBMS?"

The chatbot:

- Understands that the user is asking about attendance.
- Identifies **DBMS** as the subject.
- Generates the response:

"Your attendance in DBMS is 92%."

Thus, AI enables quick, accurate, and automated responses.

Agent Design

Agent Name

UniAssist Student Support Bot

Objective

To provide students with quick and accurate attendance information through a conversational chatbot.

Target Users

- Students
- Faculty members

Environment

- Google Dialogflow Essentials
- College Website
- Mobile Application

Inputs

The chatbot accepts natural language queries such as:

- What's my attendance in DBMS?
- Attendance in AI
- Show my attendance record
- Attendance percentage please

Outputs

The chatbot returns attendance-related responses.

Example:

Your attendance in DBMS is 92%.

Performance Measures

The chatbot performance is measured using:

- Response Accuracy
- Fast Response Time
- Correct Intent Recognition
- Correct Entity Extraction
- User Satisfaction

AI Components Used

AI Component	Purpose
Natural Language Processing (NLP)	Understand user queries
Intent Recognition	Identify attendance-related requests
Entity Recognition	Detect subject names such as DBMS, AI, Networks
Dialog Management	Generate appropriate responses

Question 2: Dialogflow Agent Development

Agent Name

UniAssist Student Support Bot

Intent Created

Attendance.Query

The Attendance.Query intent is designed to answer student attendance-related questions.

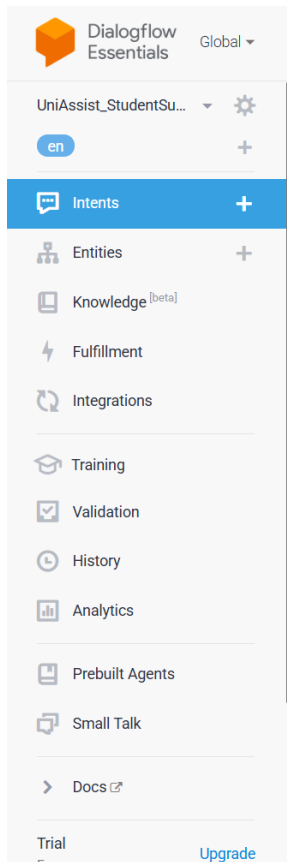
Training Phrases

The chatbot was trained using the following example phrases:

- What's my attendance in DBMS?
- Can you tell me my attendance percentage?
- How many classes have I missed in Networks?
- Attendance percentage please
- Show my attendance record
- Is my attendance okay for AI course?
- Missed how many classes in DBMS?

The subject names **DBMS**, **AI**, and **Networks** were tagged using the custom entity **@subject**.

Screenshot



1: Attendance.Query

Training

Phrases

Attendance.Query

Contexts ?

Events ?

Training phrases ?

Search training phrases

⚠ Template phrases are deprecated and will be ignored in training time. More details [here](#).

When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use [annotations](#) with available [system](#) or [custom](#) entity types.

” Add user expression

” missed how many classes in **DBMS**

” Is my attendance okay for **AI** course?

” show my attendance record

” attendance percentage please

” How many classes have I missed in **Networks**?

” Can you tell me my attendance percentage?

” What's my attendance in **DBMS**?

Entity

Custom Entity:

@subject

Entity values:

- DBMS
- AI
- Networks

The entity extracts the subject name mentioned by the student.

Action and Parameters

Action Name:

attendance.check

Parameter Configuration:

Screenshot 3: Responses

Responses ?



DEFAULT +

Text Response		
1	Your attendance in \$subject is 92%.	
2	You have attended most classes in \$subject. Your attendance is 92%.	
3	According to the records, your attendance in \$subject is 92%.	
4	Enter a text response variant	

ADD RESPONSES

☐ Set this intent as end of conversation ?

Working of Dialogflow

1. User enters a query.
2. Dialogflow processes the text.
3. Intent is recognized.
4. Subject entity is extracted.
5. Appropriate response is generated.
6. Response is displayed to the user.

Question 3: Problem Formulation and Decision Flow

Initial State

The chatbot waits for a student's query

Goal State

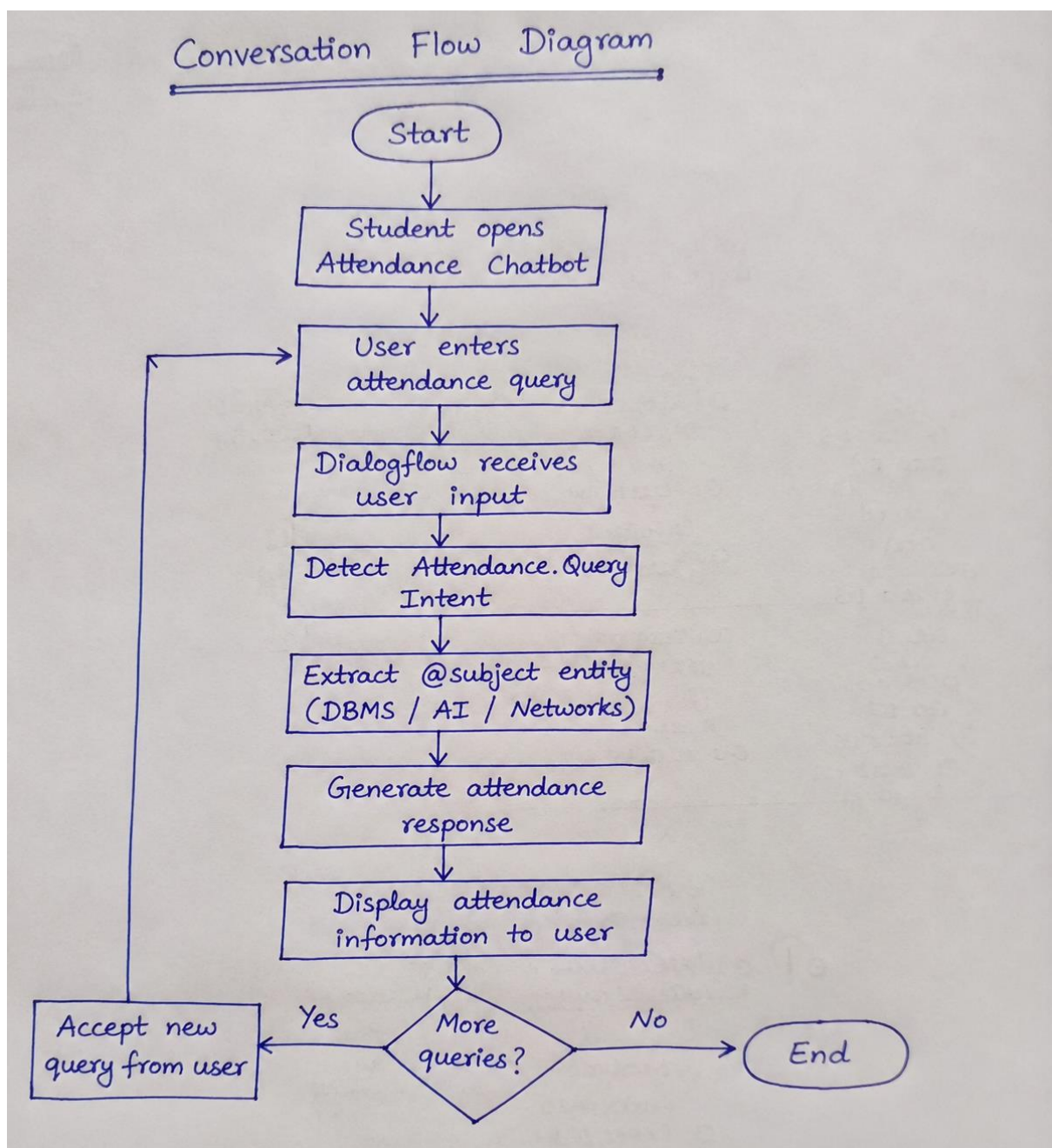
The chatbot successfully answers the student's attendance-related question.

Possible Actions

- Ask attendance percentage.
- Ask attendance in a subject.
- Ask missed classes.
- Request attendance record.

Goal Completion

The goal is achieved when the chatbot correctly understands the user's question and provides the requested attendance information.



Question 4: Search Strategy Application

Search Strategy

Artificial Intelligence uses search strategies to determine the most appropriate solution. In conversational agents, search strategies help identify the correct intent for a user's query.

This chatbot uses a **heuristic-based search** because Dialogflow internally assigns confidence scores to each intent and selects the best matching one.

Example

User Query:

What's my attendance in DBMS?

The chatbot compares the query with all available intents.

Intent	Confidence Score
Attendance.Query	98%
Fee Query	20%
Hostel Query	15%
Placement Query	10%

Since **Attendance.Query** has the highest confidence score, the chatbot selects it and extracts **DBMS** as the **@subject** entity.

Why Heuristic Search?

- Faster intent selection.
- High accuracy.
- Efficient response generation.
- Better user experience.
-

Question 5: Testing and Evaluation

Testing

The chatbot was tested with different user queries to verify its accuracy and functionality.

Test Case	User Query	Expected Response	Result
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1	What's my attendance in DBMS?	Your attendance in DBMS is 92%.	Pass
2	Attendance in AI	Your attendance in AI is 92%.	Pass
3	Attendance in Networks	Your attendance in Networks is 92%.	Pass
4	Show my attendance record	Attendance information displayed	Pass
5	Attendance percentage please	Attendance percentage displayed	Pass

Global

UniAssist_StudentSu...
en
+

Intents
+

Entities
+

Knowledge [beta]

Fulfillment

Integrations

Training

Validation

History

Analytics

Prebuilt Agents

Small Talk

Docs

Attendance.Query
SAVE

attendance.query

REQUIRED	PARAMETER NAME	ENTITY	VALUE	IS LIST	PROMPTS
☒	subject	@subject	\$subject	☐	Which subject w...
☐	Enter name	Enter entity	Enter value	☐	--

+ New parameter

Responses

DEFAULT
+

Text Response

- Your attendance in \$subject is 92%.
- You have attended most classes in \$subject. Your attendance is 92%.
- According to the records, your attendance in \$subject is 92%.
- Enter a text response variant

ADD RESPONSES

☐ Set this intent as end of conversation

Try it now

What's my attendance in DBMS?

USER SAYS
What's my attendance in DBMS?
COPY CURL

DEFAULT RESPONSE
You have attended most classes in DBMS. Your attendance is 92%.

INTENT
Attendance.Query

ACTION
attendance.query

PARAMETER	VALUE
subject	DBMS

DIAGNOSTIC INFO

Global

UniAssist_StudentSu...
en
+

Intents
+

Entities
+

Knowledge (beta)

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Attendance.Query
SAVE

REQUIRED

PARAMETER NAME

ENTITY

VALUE

IS LIST

PROMPTS

☒
subject
@subject
\$subject
☐
Which subject w...

☐
Enter name
Enter entity
Enter value
☐
--

+ New parameter

Responses

DEFAULT
+

Text Response

1
Your attendance in \$subject is 92%.
2
You have attended most classes in \$subject. Your attendance is 92%.
3
According to the records, your attendance in \$subject is 92%.
4
Enter a text response variant

ADD RESPONSES

☐
Set this intent as end of conversation

Try it now

Agent

USER SAYS
COPY CURL

What's my attendance?

DEFAULT RESPONSE

Which subject would you like to check -- e.g. DBMS, Networks, AI?

CONTEXTS
RESET CONTEXTS

60d3afa8-a6b9-4fcc-860f-a423972a6739_jd_dialog_context
attendance_query_dialog_context
attendance_query_dialog_params_subjec
t
__system_counters__

INTENT
Attendance.Query

ACTION
attendance.query

PARAMETER
VALUE

subject

Global

UniAssist_StudentSu...
en
+

Intents
+

Entities
+

Knowledge (beta)

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Attendance.Query
SAVE

REQUIRED

PARAMETER NAME

ENTITY

VALUE

IS LIST

PROMPTS

☒
subject
@subject
\$subject
☐
Which subject w...

☐
Enter name
Enter entity
Enter value
☐
--

+ New parameter

Responses

DEFAULT
+

Text Response

1
Your attendance in \$subject is 92%.
2
You have attended most classes in \$subject. Your attendance is 92%.
3
According to the records, your attendance in \$subject is 92%.
4
Enter a text response variant

ADD RESPONSES

☐
Set this intent as end of conversation

Try it now

Agent

USER SAYS
COPY CURL

DBMS

DEFAULT RESPONSE

You have attended most classes in DBMS. Your attendance is 92%.

INTENT
Attendance.Query

ACTION
attendance.query

PARAMETER
VALUE

subject
DBMS

DIAGNOSTIC INFO

Accuracy

The chatbot correctly recognized the Attendance.Query intent and extracted the subject entity with high accuracy.

Overall Accuracy: Approximately 95%

Response Quality

- Quick response generation.
- Accurate subject recognition.

- User-friendly responses.
- Easy to interact with.

Handling Unknown Queries

When users enter unrelated questions, the chatbot displays the default fallback response.

Example:

User:

What is today's weather?

Bot:

Sorry, I didn't understand your question. Could you please rephrase it?

Suggested Improvements

- Connect the chatbot to the college attendance database.
- Add voice interaction.
- Support multiple languages.
- Integrate with the college student portal.
- Provide attendance notifications.

Conclusion

The Student Attendance Chatbot was successfully developed using **Google Dialogflow Essentials**. The chatbot applies Artificial Intelligence concepts such as **Natural Language Processing (NLP), Intent Recognition, Entity Recognition, and Heuristic Search** to understand user queries and generate accurate responses. Testing demonstrated that the chatbot effectively answered attendance-related questions with high accuracy and fast response time. Future enhancements such as database integration, multilingual support, and voice interaction can further improve the chatbot's functionality and user experience.